Do Now · retrieval

Lesson 140 · Referential Integrity, Indexing and Keys

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

What is the result of UPDATE without a WHERE clause?

Q2

What is a trace table used for?

Q3

How many bits are in one byte?

Q4

What data structure uses LIFO?

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Link the records — Order.CustomerID references Customer.CustomerID.

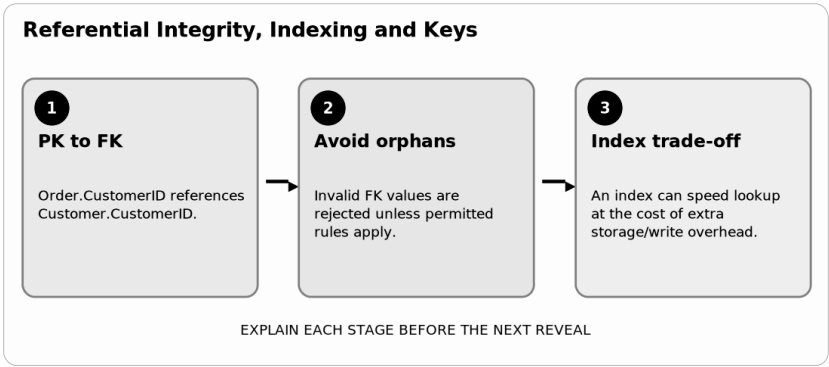
Core idea 2

Enforce integrity — Invalid FK values are rejected unless permitted rules apply.

Core idea 3

Improve searches — An index can speed lookup at the cost of extra storage/write overhead.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

referential integrity — Foreign-key values correspond to existing referenced keys or permitted NULLs.

index — Auxiliary structure helping the DBMS find rows efficiently.

primary key — Unique identifier of each record.

foreign key — Attribute referencing another relation.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

Problem

Customer has IDs 7 and 9. A new Order has CustomerID 12. Explain what happens under enforced referential integrity.

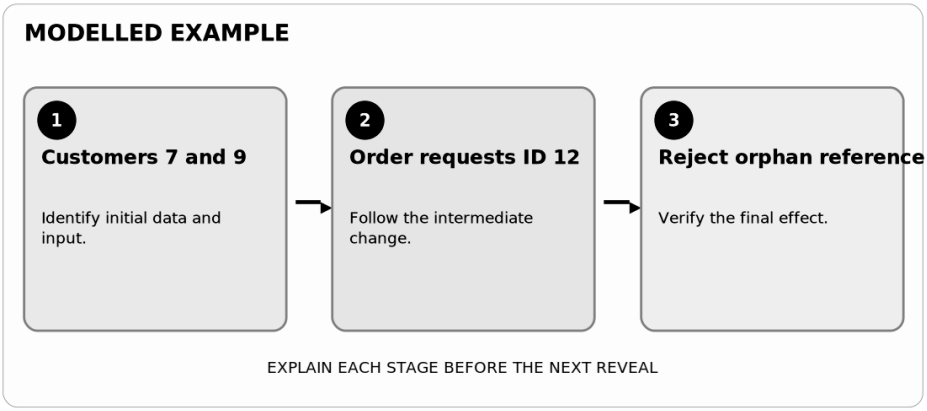
Follow the worked method

1. Customers 7 and 9 — Identify the relevant information and make a prediction.

2. Order requests ID 12 — Explain the change using correct technical terms.

3. Reject orphan reference — The DBMS rejects the foreign key value 12, unless a matching Customer row is created or the reference is changed. Policies for deleting parent rows may RESTRICT, CASCADE or SET NULL, where permitted and configured.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Frequent ISBN lookup
Identify the relevant information and make a prediction.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Explain referential integrity using Borrower and Loan with a suitable example. [6]

Question 2

Identify two benefits and one cost of indexing a frequently searched field. [6]

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

A library deletes Borrower ID 8 while Loan rows refer to 8. Compare RESTRICT and CASCADE and justify a safe policy. [8]

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

An index guarantees every column value it indexes is unique.

Use key vocabulary from memory

1. Explain referential integrity in your own words.

2. Explain index in your own words.

3. Explain primary key in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.